

MARKOV DYNAMIC PROGRAMMING

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A TYPICAL PROBLEM

- The planner chooses a path of actions $(A_t)_{t \geq 0}$ to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t r(X_t, A_t)$$

where $(X_t)_{t \geq 0}$ is a state process (X_0 is given).

- X is a finite set: **state space**.
- A is a finite set: **action space**.
- Γ is a **correspondence** from X to A . Intuitively: the set of actions feasible given the state.

MDP

- Given A and X , a finite **Markov decision process** (MDP) is a tuple $\mathcal{M} = (\Gamma, P, r, \beta)$ where
 - $\Gamma : X \rightarrow A$ is a nonempty correspondence from X to A defining feasible state-action pairs

$$G := \{(x, a) \in X \times A : a \in \Gamma(x)\}$$

- a **stochastic kernel** P from G to X :

$$\sum_{x' \in X} P(x, a, x') = 1 \text{ for all } (x, a) \in G$$

- a function r from G to $\mathbb{R}$ is a **reward function**
- $\beta \in (0, 1)$ is a discount factor

MDP

- The Bellman equation associated with $\mathcal{M}$ is

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \sum_{x' \in X} P(x, a, x') v(x') \right\} \text{ for all } x \in X.$$

- This is an equation in the unknown function $v \in \mathbb{R}^X$ ($\mathbb{R}^X$ is the set of all functions from X to $\mathbb{R}$).
- We will show that the solution to the Bellman equation equals the largest possible value of the objective function in the [sequence problem](#):

$$\max \mathbb{E} \sum_{t=0}^{\infty} \beta^t r(X_t, A_t), \quad \text{subject to } A_t \in \Gamma(X_t) \text{ for all } t \geq 0.$$

POLICIES

- Let Σ be the set of all **feasible policies** given $\mathcal{M}$:

$$\Sigma := \left\{ \sigma \in \mathcal{A}^X : \sigma(x) \in \Gamma(x) \text{ for all } x \in X \right\}.$$

- For any $\sigma \in \Sigma$ we have P_σ is a stochastic kernel from X to X :

$$P_\sigma(x, x') := P(x, \sigma(x), x') \text{ for all } (x, x') \in X \quad \text{so } P_\sigma \in \mathcal{M}(\mathbb{R}^X).$$

Note: notational issue - $\mathcal{M}$ here is not MDP, it is a set of Markov operators.

- Similarly, for any $\sigma \in \Sigma$ we have r_σ , a function from X to $\mathbb{R}$:

$$r_\sigma(x) := r(x, \sigma(x)) \text{ for all } x \in X \quad \text{so } r_\sigma \in \mathbb{R}^X.$$

POLICIES

- Define $\mathbb{E}_{x_0} [\cdot] := \mathbb{E} [\cdot \mid X_0 = x_0]$. The **lifetime value** of following $\sigma \in \Sigma$ from x is

$$v_\sigma(x) := \mathbb{E}_x \left[\sum_{t=0}^{\infty} \beta^t r_\sigma(X_t) \right]$$

where X_t is P_σ -Markov with $X_0 = x$.

- Since $\beta \in (0, 1)$, we can calculate

$$v_\sigma(x) = \sum_{t=0}^{\infty} \beta^t P_\sigma^t r_\sigma = (I - \beta P_\sigma)^{-1} r_\sigma.$$

POLICY OPERATOR

- Define the **policy operator** T_σ :

$$(T_\sigma v)(x) := r(x, \sigma(x)) + \beta \sum_{x' \in X} v(x') P(x, \sigma(x), x') \text{ for all } x \in X.$$

- We denote a fixed point of T_σ by v_σ .
- We will now prove T_σ is a contraction of modulus β on $\mathbb{R}^X$ under norm $\|\cdot\|_\infty$.
- We will also show that T_σ is **order-preserving**: if $v \leq w$ then $T_\sigma v \leq T_\sigma w$.

POLICY OPERATOR

- Take any $v, w \in \mathbb{R}^X$ and $\sigma \in \Sigma$.
- Fix $x \in X$. We have

$$\begin{aligned} |(T_\sigma v)(x) - (T_\sigma w)(x)| &= \beta \left| \sum_{x' \in X} (v(x') - w(x')) P(x, \sigma(x), x') \right| \\ &\leq \beta \sum_{x' \in X} |v(x') - w(x')| P(x, \sigma(x), x') \\ &\leq \beta \|v - w\|_\infty \end{aligned}$$

- Since this is true regardless of x , we have

$$\|T_\sigma v - T_\sigma w\|_\infty \leq \beta \|v - w\|_\infty .$$

POLICY OPERATOR

- To show that T_σ is order preserving, take any $v, w \in \mathbb{R}^X$ with $v \leq w$ and $\sigma \in \Sigma$.
- $v \leq w$ implies $P_\sigma v \leq P_\sigma w$. We can write

$$T_\sigma v = r_\sigma + \beta P_\sigma v \text{ and } T_\sigma w = r_\sigma + \beta P_\sigma w,$$

so $T_\sigma v \leq T_\sigma w$.

GREEDY POLICIES

- Given MDP $\mathcal{M}$ the **value function** is

$$v^*(x) := \max_{\sigma \in \Sigma} v_{\sigma}(x) \text{ for all } x \in X.$$

- We call a policy $\sigma \in \Sigma$ **optimal** if $v_{\sigma} = v^*$.
- We call a policy **v-greedy** if

$$\sigma(x) \in \operatorname{argmax}_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \sum_{x' \in X} v(x') P(x, a, x') \right\} \text{ for all } x \in X.$$

BELLMAN OPERATOR

- We say that **Bellman's principle of optimality** holds for MDP $\mathcal{M}$ if

$$\sigma \in \Sigma \text{ is optimal for } \mathcal{M} \iff \sigma \text{ is } v^* \text{-greedy.}$$

- The **Bellman operator** corresponding to $\mathcal{M}$ is a self-map T on $\mathbb{R}^X$ defined by

$$Tv(x) := \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \sum_{x' \in X} v(x') P(x, a, x') \right\} \text{ for all } x \in X.$$

OPTIMALITY

Theorem

Let $\mathcal{M}$ be an MDP with Bellman operator T . Then

- 1. v^* is the unique solution to the Bellman equation $v = Tv$ in $\mathbb{R}^X$,*
- 2. $\lim_{k \rightarrow \infty} T^k v = v^*$ for all $v \in \mathbb{R}^X$,*
- 3. Bellman's principle of optimality holds for $\mathcal{M}$,*
- 4. at least one optimal policy exists.*

OPTIMALITY

- Instead of solving the (possibly hard) **sequence problem** we can solve the (possibly easier) **functional equation** $v = Tv$.
- Finding v -greedy policies is easier than looking at the entire set of feasible policies Σ .
- The required conditions are pretty weak. Important and somewhat hidden: sets are finite and $r : G \rightarrow \mathbb{R}$.

OPTIMALITY

- We will prove (1) and (2).
- Two parts of the proof:
 1. Show there exists a unique fixed point of T .
 2. Show that the fixed point is v^* .

PROOF: T IS A CONTRACTION

- Fix v, w in $\mathbb{R}^X$. For any x , we have

$$\begin{aligned} |(Tv)(x) - (Tw)(x)| &= \left| \max_{a \in \Gamma(x)} (T_a v)(x) - \max_{a \in \Gamma(x)} (T_a w)(x) \right| \\ &\leq \max_{a \in \Gamma(x)} |(T_a v)(x) - (T_a w)(x)| \end{aligned}$$

where T_a denotes the policy operator for any policy σ with $\sigma(x) = a$.

- Since $|(T_a v)(x) - (T_a w)(x)| \leq \|T_a v - T_a w\|_\infty$, we have

$$\|Tv - Tw\|_\infty \leq \max_{\sigma \in \Sigma} \|T_\sigma v - T_\sigma w\|_\infty.$$

PROOF: T IS A CONTRACTION (CONT.)

- We showed earlier that T_σ is a **contraction**:

$$\|T_\sigma v - T_\sigma w\|_\infty \leq \beta \|v - w\|_\infty.$$

- We thus have

$$\|Tv - Tw\|_\infty \leq \beta \|v - w\|_\infty \text{ for all } v, w \in \mathbb{R}^X.$$

- By the **Banach fixed point theorem** T has a **unique fixed point** $\bar{v}$.

PROOF: FIXED POINT IS OPTIMAL

- We will now show that $\bar{v} = v^*$.
- **Step 1:** $\bar{v} \leq v^*$.
 - Pick $\sigma \in \Sigma$ that is $\bar{v}$ -greedy.
 - By definition $T_\sigma \bar{v} = T \bar{v} = \bar{v}$.
 - So $\bar{v}$ is a fixed point of T_σ , which implies $\bar{v} = v_\sigma$.
 - Since $v^* := \max_{\sigma'} v_{\sigma'}$, we have $\bar{v} \leq v^*$.
- **Step 2:** $v^* \leq \bar{v}$.
 - Pick any $\sigma \in \Sigma$. Note that $T_\sigma v \leq T v$ for any v .
 - In particular, $T_\sigma \bar{v} \leq T \bar{v} = \bar{v}$.
 - Since T_σ is order preserving, applying it repeatedly yields $T_\sigma^k \bar{v} \leq \bar{v}$.
 - Taking the limit $k \rightarrow \infty$, we get $v_\sigma \leq \bar{v}$.
 - Since this holds for any σ , $v^* \leq \bar{v}$.

PROOF: BELLMAN PRINCIPLE OF OPTIMALITY

- We want to show: σ is optimal $\iff \sigma$ is v^* -greedy.
- ($\Rightarrow$) Let σ be optimal, so $v_\sigma = v^*$.
 - We know v^* is the fixed point of T , so $v^* = Tv^*$.
 - We know v_σ is the fixed point of T_σ , so $v_\sigma = T_\sigma v_\sigma$.
 - Thus $T_\sigma v^* = v^* = Tv^*$.
 - By definition of T , this implies $\sigma(x)$ maximizes the RHS of Bellman equation, so σ is v^* -greedy.
- ($\Leftarrow$) Let σ be v^* -greedy.
 - By definition of greedy, $T_\sigma v^* = Tv^*$.
 - Since v^* is the fixed point of T , $Tv^* = v^*$.
 - So $T_\sigma v^* = v^*$, meaning v^* is a fixed point of T_σ .
 - Since T_σ has a unique fixed point v_σ , we have $v_\sigma = v^*$. Thus σ is optimal.

POLICY ITERATION

- We can use T_σ to look for the value function (instead of value function iteration).
- Start with a guess v_0 , find a greedy policy σ_0 and calculate the fixed point of T_{σ_0} :

$$v_{\sigma_0} = (I - \beta P_{\sigma_0})^{-1} r_{\sigma_0}.$$

- Repeat the process with v_{σ_0} – find a greedy policy and calculate the new fixed point.
- Do it until convergence.
- This algorithm is known as **policy iteration** or **Howard's policy iteration**

HPI

Algorithm Howard's Policy Iteration

```
1: procedure HPI
2:    $k \leftarrow 1, \epsilon \leftarrow \tau + 1, v_k \leftarrow v_{\text{init}}$ 
3:   while  $\epsilon > \tau$  do
4:      $\sigma_k \leftarrow v_k$ -greedy policy
5:      $v_{k+1} = (I - \beta P_{\sigma_k})^{-1} r_{\sigma_k}$ 
6:      $\epsilon \leftarrow \|v_{k+1} - v_k\|_{\infty}, k \leftarrow k + 1$ 
7:   end while
8: end procedure
```

HPI

- HPI converges at a faster rate than VFI.
- In a finite state setting, the algorithm always converges to an exact optimal policy in a finite number of steps, regardless of the initial condition.
- Drawback: computing v_{σ} can be expensive (requires matrix inversion).

OPTIMISTIC POLICY ITERATION

- This is a variant of HPI.
- Key difference: do not compute v_σ exactly.
- Instead, apply the policy operator T_σ to v_k for a fixed number of iterations, m .
- For $m \rightarrow \infty$ we have HPI; for $m = 1$ we have VFI.
- Often outperforms HPI and VFI, but this requires choosing m .

OPI

Algorithm Optimistic Policy Iteration

```
1: procedure OPI
2:    $k \leftarrow 1, \epsilon \leftarrow \tau + 1, v_k \leftarrow v_{\text{init}}$ 
3:   while  $\epsilon > \tau$  do
4:      $\sigma_k \leftarrow v_k$ -greedy policy
5:      $v_{k+1} = T_{\sigma_k}^m v_k$ 
6:      $\epsilon \leftarrow \|v_{k+1} - v_k\|_{\infty}, k \leftarrow k + 1$ 
7:   end while
8: end procedure
```
